Homework 1

Problem 1:

$$H(x) = -\sum_{1}^{\infty} 2^{-k} \log_2^{(2^{-k})} \qquad \log_2^{(2^{-k})} = -k.$$

$$= \sum_{1}^{\infty} k 2^{-k}.$$

$$(2^{-1}) < 1$$

則 $\sum_{1}^{\infty} k 2^{-k} = \frac{2^{-1}}{(1-2^{-1})^2} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2.$

綜上 $H(x) = 2$ bits.

Problem 2:

$$p(x) = \{\tfrac{5}{32}, \tfrac{7}{32}, \tfrac{1}{4}, \tfrac{3}{8}\}. \quad p(Y) = \{1/2, 1/4, 1/8, 1/8\}$$

$$8/32 \quad 12/32$$

$$H(X) = -\frac{5}{32}\log_2^{\frac{5}{32}} - \frac{7}{32}\log_2^{\frac{7}{32}} - \frac{1}{4}\log_2^{\frac{1}{4}} - \frac{3}{8}\log_2^{\frac{3}{8}}$$

$$\log_2^{\frac{k}{32}} = \log_2^{k} - 5$$

$$= 5 - 2^{-5}(5\log_2^{5} + 7\log_2^{7} + 24 + 12\log_2^{12}) \approx 1.93 \text{ bits}.$$

$$H(Y) = -\frac{1}{2}\log_2^{\frac{1}{2}} - \frac{1}{4}\log_2^{\frac{1}{4}} - \frac{1}{8}\log_2^{\frac{1}{8}} - 1/8\log_2^{1/8} = 1.75 \text{ bits}$$

$$H(X,Y) = -\sum p(x,y)\log_2^{p(x,y)}$$

$$= -\frac{1}{4}\log_2^{\frac{1}{4}} - \frac{1}{8}\log_2^{\frac{1}{8}} \times 2 - \frac{1}{16}\log_2^{1/16} \times 4 - \frac{1}{32}\log_2^{1/32} \times 8 = 3.5 \text{ bits}.$$

$$H(X|Y) = H(X,Y) - H(Y) = 1.75 \text{ bit}$$

$$H(Y|X) = H(X,Y) - H(X) = 1.57 \text{ bit}$$

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = 0.18$$

Problem 3.

Proof of 给定均值且 $x>0$ 下，最大熵分布是指数分布.

$$\max_{p(x)} h(p(x)) \quad s.t. \int_0^\infty p(x)\,dx = 1 \leftarrow \text{概率密度}.$$

$$\int_0^\infty x\,p(x)\,dx = \lambda \leftarrow E[x]=\lambda.$$

可将优化问题使用拉格朗日乘子改写，如下

$$\max_{p(x)} h(p(x)) = -\int_0^\infty p(x)\log_2 p(x)\,dx + \mu\left(\int_0^\infty p(x)\,dx - 1\right) + \xi\left(\int_0^\infty x\,p(x)\,dx - \lambda\right)$$

$$= -\int_0^\infty \left(p(x)\log_2 p(x) + \mu p(x) + \xi x p(x)\right) dx - \mu - \xi\lambda$$

令 $F(x,p) = -p\log_2 p + \mu p + \xi x p$

当 $\frac{\partial F}{\partial p} = 0$ 时，有极值.

为方便求导，$\log_2 p \Rightarrow \frac{\ln p}{\ln 2}$，可以提到 $h(p(x))$ 外.

与 $h(p(x))$ 在一样的点取最大值，此时

$F(x) = -p\ln p + \mu p + \xi x p$.

$\frac{\partial F}{\partial p} = -(\ln p + 1) + \mu + \xi x$.

$\ln p(x) = \mu + \xi x - 1 \Rightarrow p(x) = e^{\mu-1} e^{\xi x}$

由于 $p(x)$ 在 $[0,+\infty)$ 可积，则 $\xi<0$.

$\xi = -c$，$c>0$ $A = e^{\mu-1}$ $p(x) = Ae^{-cx}$.

因为 $\int_0^\infty Ae^{-cx}dx = 1 = \frac{A}{c}$ $\therefore A = c$.

$p(x) = c e^{-cx}$.

又 $\because \int_0^\infty x c e^{-cx} dx = \frac{1}{c} = \lambda \Rightarrow c = \frac{1}{\lambda}$

综上 $p(x) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}$ 有最大熵，代入 $h(p(x))$

$$h(p(x)) = -\int_0^\infty \frac{1}{\lambda} e^{-\frac{x}{\lambda}} \log_2 \frac{1}{\lambda} e^{-\frac{x}{\lambda}} dx$$

$$= -\int_0^\infty \frac{1}{\lambda} e^{-\frac{x}{\lambda}} \left(-\log_2 \lambda - \frac{x}{\lambda} \log_2 e\right) dx$$

$$= \int_0^\infty \frac{1}{\lambda} e^{-\frac{x}{\lambda}} \left(\log_2 \lambda + \frac{x}{\lambda} \log_2 e\right) dx$$

$$= \log_2(e\lambda)$$

综上，给定均值 λ，且 $x>0$ 时，在 $p(x) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}$ 有最大熵 $\log_2(e\lambda)$.

Problem 4:

对于 $X = Y^3$，$H(X) = H(Y)$，因为 X 被 $\sqrt[3]{\ }$ 一一映射到 Y.

对于 $X = Y^2$，$H(X) \geq H(Y)$，在 $X \geq 0 / X \leq 0$ 时取等号.

因为 $\sqrt{\ }$ 把 $x, -x$ 映射到 x^2

对于 $Y = g(X)$，Y 无不确定性（知道 X 时）$H(Y|X) = 0$，

$$H(X,Y) = H(X) + H(Y|X) = H(X)$$

$$= H(Y) + H(Y|X)$$

所以 $H(X) = H(Y) + H(Y|X)$

由于条件熵 $H(Y|X) \geq 0$，

所以 $H(X) \geq H(Y) \Rightarrow \underline{H(g(X)) \leq H(X)}$